

Multi-platform distributed systems with Aggregate Computing in Kotlin: the case of proximity messaging

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Overview and Motivations

- **Context:** The world is increasingly populated by IoT and mobile devices that must communicate in dynamic environments.
- **Problem:** Centralized messaging architectures are fragile, difficult to scale, and unsuitable for pervasive and ubiquitous computing scenarios.
- **Motivation:** Resilient and more adaptive communication is needed.

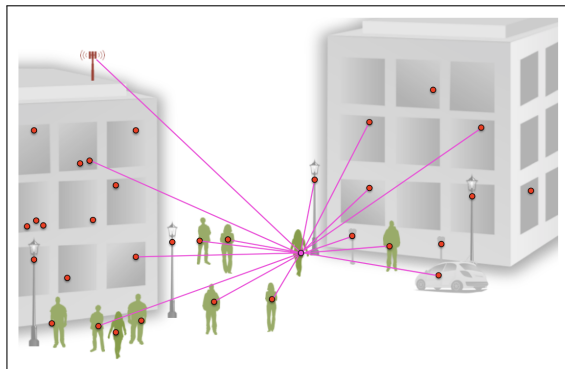


Figure: Example of a centralized system for proximity messaging.

Goals

This thesis project has two main goals:

Goal 1

Design and implement a decentralized, proximity-based messaging algorithm based on the principles of Aggregate Computing.

Goal 2

Develop a multiplatform (Android/iOS) mobile application that uses the algorithm as core logic.

Aggregate Computing: The paradigm

What is Aggregate Computing?

Aggregate Computing

Programming paradigm that treats large, interconnected systems of devices as a single, aggregate machine rather than focusing on individual components.

Key concepts

- **Collective behavior:** The focus is on programming the group, or "aggregate," as a whole.
- **Abstraction:** It provides a higher level of abstraction that hides the complexity of managing individual devices in a vast and dynamic network.
- **Logic:** The program logic is expressed as the manipulation of data structures that extend spatially and temporally (**Field Calculus**).

Architecture

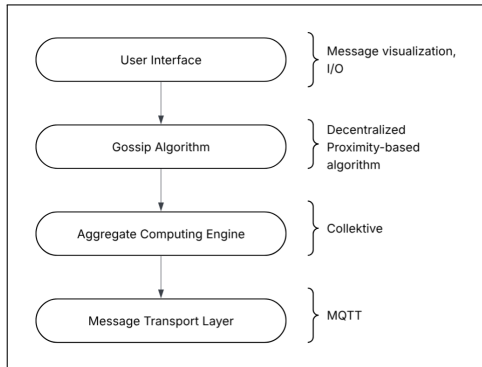


Figure: High-level overview of the architecture.

Technologies

- **AC framework:** Kollektive
- **Programming language:** Kotlin
- **Multi-platform framework:** KMP
- **GUI:** Compose Multiplatform
- **Simulator:** Alchemist Simulator
- **Transport Protocol:** MQTT

Algorithm (Gossip-Gradient)

The algorithm is at the core of the system, allowing decentralized proximity messaging.

Description

- **Objective:** Enable communication between nodes by finding the shortest path (**gradient**) to the message source.
- **Algorithm Family:** **Gossip-based** algorithm, which rely on local interactions between nodes to propagate information throughout the network. It uses a **Bellman-Ford-like** approach to compute the gradient.
- **Multiple Sources:** The algorithm can handle multiple message sources, allowing nodes to receive messages from different origins.
- **Parameters:** Maximum propagation distance, expiration time (TTL).

- **Simulations:** The correctness of the algorithm has been validated and tested using the Alchemist Simulator.
- **Continuous Integration:** GitHub Actions is used as CI platform to automate the build processes for both the shared codebase and the mobile application.

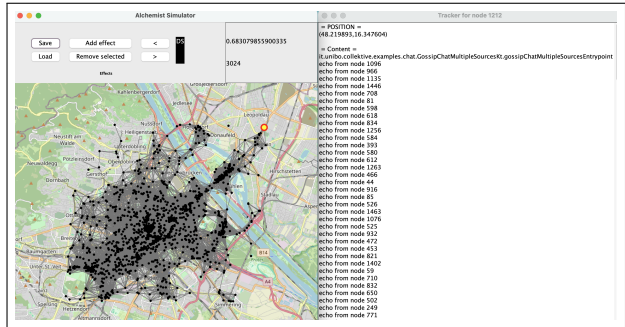


Figure: Simulation of the messaging system using real-world GPS traces. Image taken from Alchemist Simulator illustrating nodes and their communication links.

Mobile Application (Echo)

The mobile application serves as the **GUI** of the messaging system.

Features

- **Cross-platform:** KMP ensures compatibility with both Android and iOS devices.
- **AC runtime environment:** Kollektive Engine.
- **GPS:** used to compute distances between devices.
- **MQTT:** used as transport protocol for message dissemination.

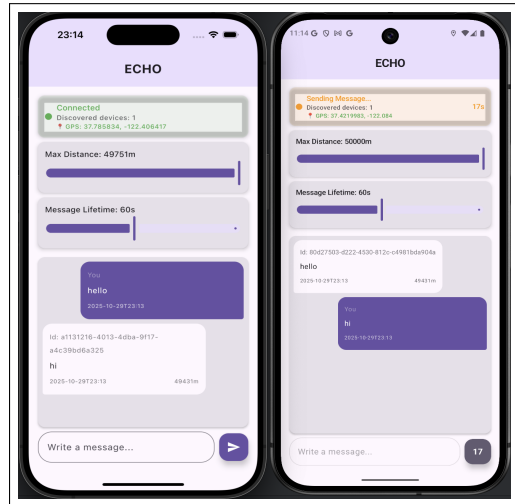


Figure: Illustration of the Echo application interface.

Conclusions

Final considerations

A decentralized messaging system based on the principles of Aggregate Computing has been successfully implemented, integrating the paradigm into a multi-platform mobile application.

Future development

- **Fully decentralization:** Replace MQTT with a P2P protocol like Bluetooth or Wi-Fi Direct.
- **Deployment:** Publish Echo on official app stores to test the application in real-world scenarios.

The End